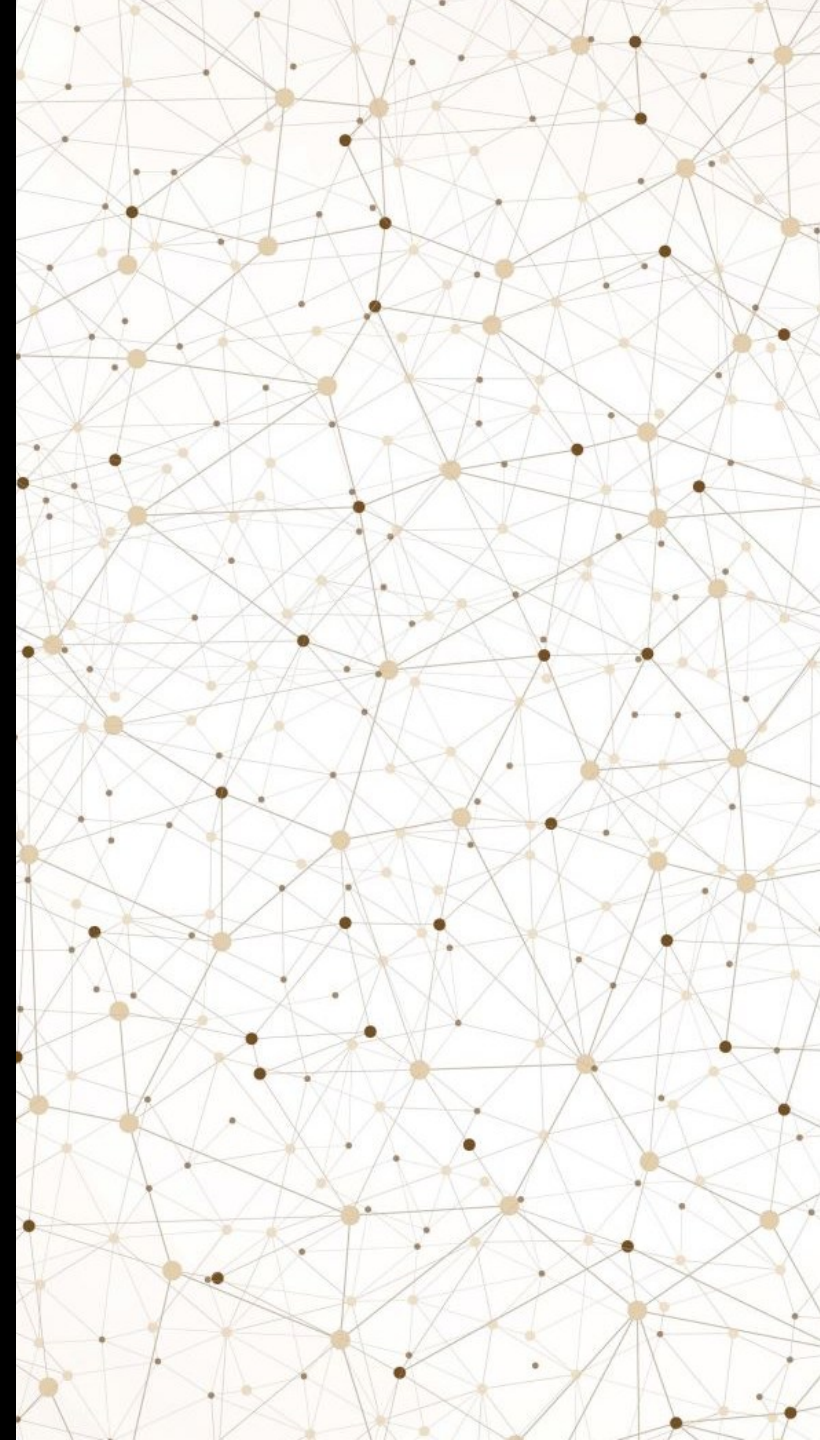


Amazon Delivery Project Documentation

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Goal project:

- The objective of this project is to enhance delivery speed by analyzing key factors influencing performance. The analysis aims to provide actionable insights that support faster, more efficient, and reliable deliveries.



Brief About The Dataset:

- This dataset contains delivery orders from February, March, and April 2022. It includes details about the delivery agent, store and drop locations, and order timings. External factors such as weather and traffic conditions are also recorded. Finally, it tracks delivery time and product category, making it useful for analysis and service improvement:



Column Name	Data Type	Description
Order_ID	String	The age of the delivery agent.
Agent_Age	Integer	The rating of the delivery agent, likely on a scale from 1 to 5.
Agent_Rating	Decimal	The latitude of the store from which the order is picked up.
Store_Latitude	Decimal	The longitude of the store from which the order is picked up.
Store_Longitude	Decimal	The latitude of the delivery location. Drop_Longitude: The
Drop_Latitude	Decimal	The latitude of the delivery location.
Drop_Longitude	Decimal	The longitude of the delivery location.
Order_Date	Date	The date when the order was placed.
Order_Time	Time	The time when the order was placed.
Pickup_Time	Time	The time when the order was picked up by the delivery agent.
Weather	String	The weather conditions at the time of delivery (e.g., Sunny, Stormy, Cloudy).
Traffic	String	The traffic conditions during the delivery (e.g., High, Medium, Low).
Vehicle	String	The type of vehicle used for delivery (e.g., motorcycle, scooter, van).
Area	String	The type of area where the delivery is made (e.g., Urban, Metropolitan).
Delivery_Time	Time	The time taken to deliver the order, likely measured in minutes.
Category	String	The category of the item being delivered (e.g., Clothing, Electronics, Toys).

- **About Dataset**

- This Amazon Delivery Dataset provides a comprehensive view of the company's last-mile logistics operations. It includes data on over 43,632 deliveries across multiple cities, with detailed information on order details, delivery agents, weather and traffic conditions, and delivery performance metrics. The dataset enables researchers and analysts to uncover insights into factors influencing delivery efficiency, identify areas for optimization, and explore the impact of various variables on the overall customer experience.

Dataset Issues:

Order_ID:

- **Description:** Unique identifier for each order.
- **Issues:** No missing values, but values are non-numeric (strings).
- **Handling:** Kept as is since it is just an identifier.
- **Impact:** Used as a reference key, not for analysis.

Agent_Age:

Description: Age of the delivery agent.

Issues: Possible invalid values (e.g., <18 or >70).

Handling: Outliers filtered and replaced with median age.

Impact: Ensures fair analysis of age-related performance without skew.

Agent_Rating:

- **Description:** Customer rating of the delivery agent (1–5).
- **Issues:** 54 missing values.
- **Handling :** Missing ratings replaced with mean/median rating.
- **Impact:** Prevents bias in analyzing performance vs. ratings.

Store_Latitude & Store_Longitude:

- **Description:** Location of the store (pickup point).
- **Issues:** No nulls but long names.
- **Handling:** Renamed to Store_Lat, Store_Lon.
- **Impact:** Used for calculating delivery distance

Drop_Latitude & Drop_Longitude:

- **Description:** Location of the customer (drop point).
- **Issues:** No nulls, same naming issue.
- **Handling:** Renamed to Drop_Lat, Drop_Lon.
- **Impact:** Paired with store location to calculate delivery distance

Dataset Issues:



Weather:

- **Description:** Weather condition during delivery.
- **Issues:** 91 missing values.
- **Handling:** Missing values replaced with "Unknown". Categories grouped into Good_Weather, Bad_Weather, Extreme.
- **Impact:** Helped show delivery delays in stormy/sandstorm conditions.

Traffic:

- **Description:** Traffic level during delivery (Low, Medium, High, Jam).
- **Issues:** Inconsistent spelling issues (standardized).
- **Handling:** Cleaned and standardized categories.
- **Impact:** Strong correlation with delivery time (higher traffic → longer delivery).

Vehicle:

- **Description:** Vehicle type used (motorcycle, scooter, etc.).
- **Issues:** No nulls.
- **Handling:** Standardized text formatting.
- **Impact:** Compared delivery times between vehicles.

Area:

- **Description:** Type of delivery area (Urban, Semi-Urban, Metropolitan).
- **Issues:** Typo in "Metropolitian" → corrected to "Metropolitan".
- **Handling:** Standardized category names.
- **Impact:** Used to compare delivery speed in different regions.

Delivery_Time:

- **Description:** Actual delivery duration in minutes.
- **Issues:** Possible extreme outliers (too small or too high).
- **Handling:** Outliers capped based on IQR method.
- **Impact:** Provided reliable analysis of delivery efficiency.

Category:

- **Description:** Type of product ordered (Clothing, Electronics, etc.).
- **Issues:** No nulls. Some categories very small in count.
- **Handling:** Kept all, but grouped very rare categories into "Other".
- **Impact:** Helped analyze if product type affects delivery time.

Data Cleaning and Preprocessing :



1. Column Renaming:

Problem:

- The column Agent_Rating was not fully descriptive

Solution:

- Renamed to Order_Rating for clarity

Impact:

- Clearer understanding that rating refers to the order

Handling Missing Values:

Order_Rating:

Problem:

- Missing ratings

Solution:

- Filled with the **median rating**

Impact:

- Prevented bias and preserved data distribution

Weather:

Problem: Missing weather conditions.

Solution: Filled with the **mode value ("Fog")**.

Impact: Maintained realistic category distribution.

Traffic:

Problem: Some values stored as the string 'NaN '.

Solution: Replaced with actual NaN.

Impact: Ensured accurate handling of missing values.

•Order_Time

Problem: Invalid placeholders such as 'NaN' text.

Solution: Converted them to proper NaN.

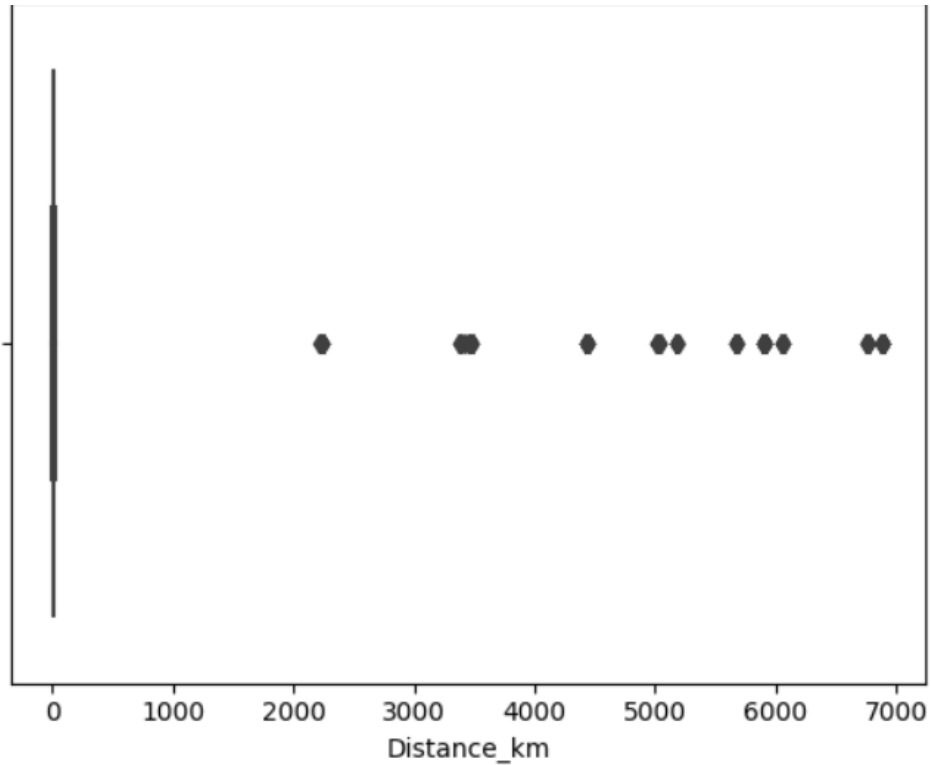
Impact: Cleaned time column for later datetime conversion.

Outlier Detection & Treatment:

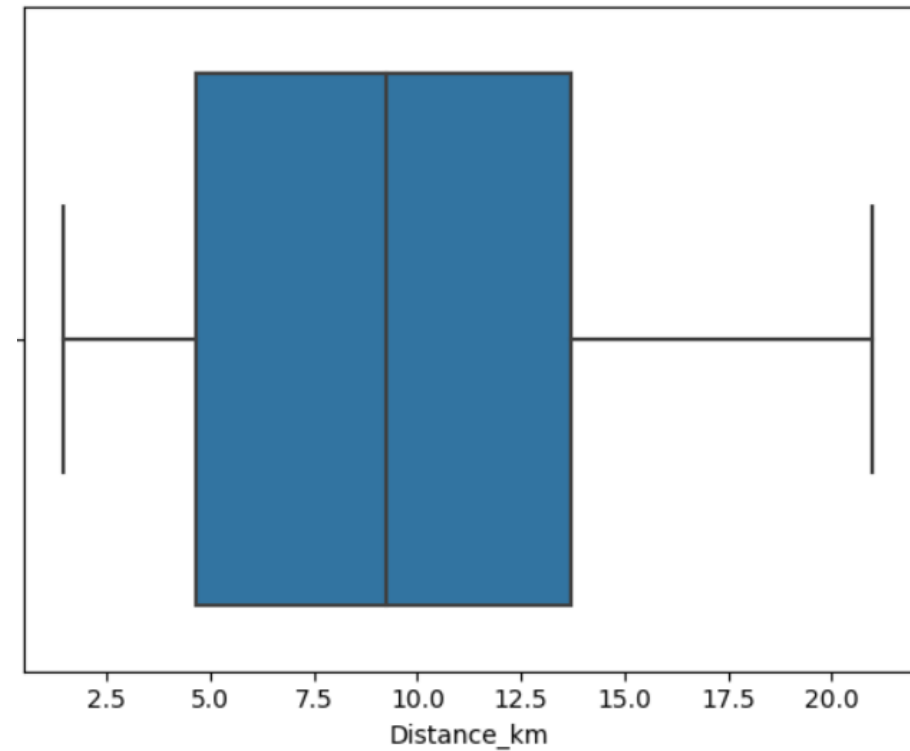
- **Agent_Age**
 - Checked distribution → detected possible outliers (e.g., very young or very old ages).
- **Order_Rating**
 - Checked distribution → ensured values within valid range (1–5).
- **Delivery_Time**
 - Distribution examined for extreme values.
- **Distance_km**
 - Used boxplot + IQR method.
 - Filtered rows outside [**lower_bound**, **upper_bound**].
- **Impact:** Outliers removed → improved accuracy of averages and predictive modeling.

Outlier Detection & Treatment:

CHECK OUTLIERS IN DISTANCE



AFTER HANDLE OUTLIER



. Feature Engineering:



Weather_Category:

Created a new column grouping weather into categories:

Good_Weather: Sunny

Bad_Weather: Cloudy, Windy, Fog

Extra_Bad_Weather: Stormy, Sandstorms

Impact: Simplified weather impact analysis on delivery.

Order_DateTime:

Combined Order_Date and Order_Time into a single datetime column.

Impact: Enabled time-based calculations (waiting time, delays, peak hours).

Order_Time_Category:

Created a new column grouping order_time into categories:(Morning, Afternoon, Evening, Night)

Area_Category : Created a new column grouping area into

Categories:(Urban, Metropolitan,Other)

- Distance_km:** Calculated using the **Haversine formula** based on pickup and drop-off GPS coordinates.

- Speed_kmph:** Computed using the formula:

$\text{Speed(km/h)} = \frac{\text{Distance_km}}{\text{Delivery_Time (minutes)} / 60}$

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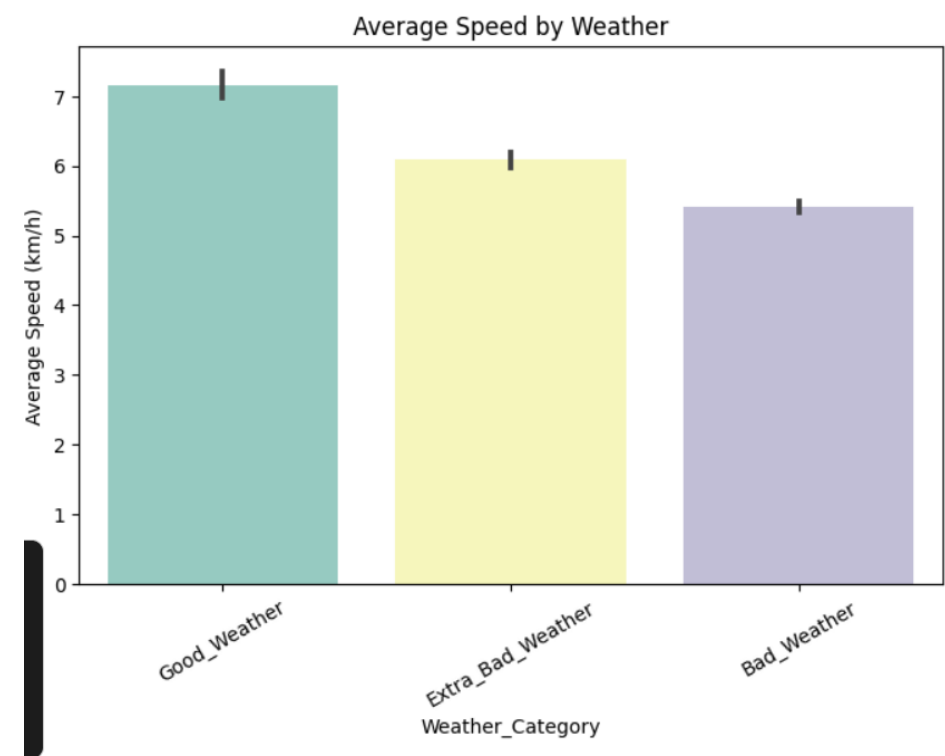
- Final dataset now contains two additional columns:

- Distance_km

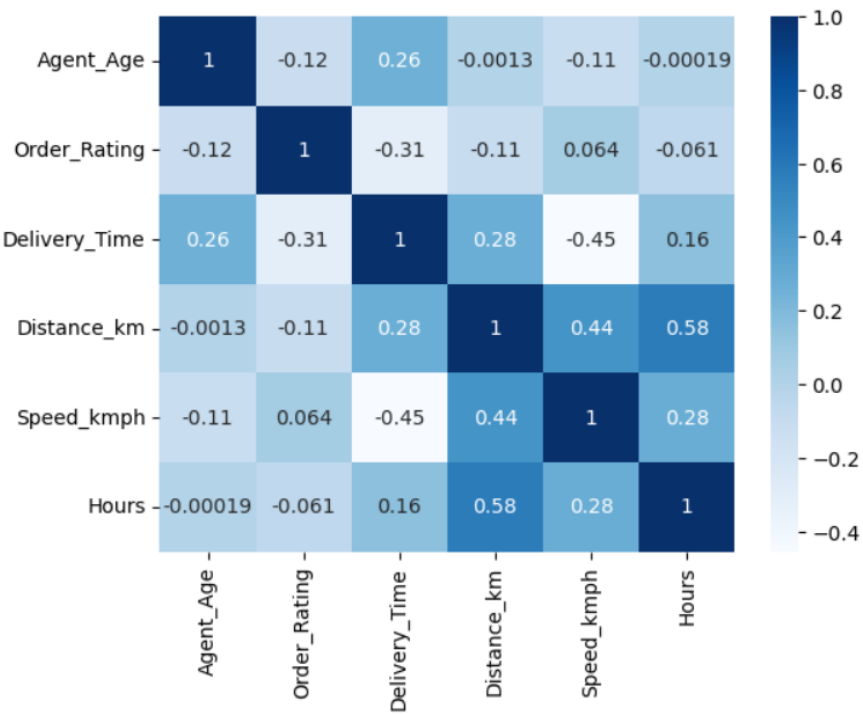
- Speed_kmph

Visualization:

Average Speed by Weather

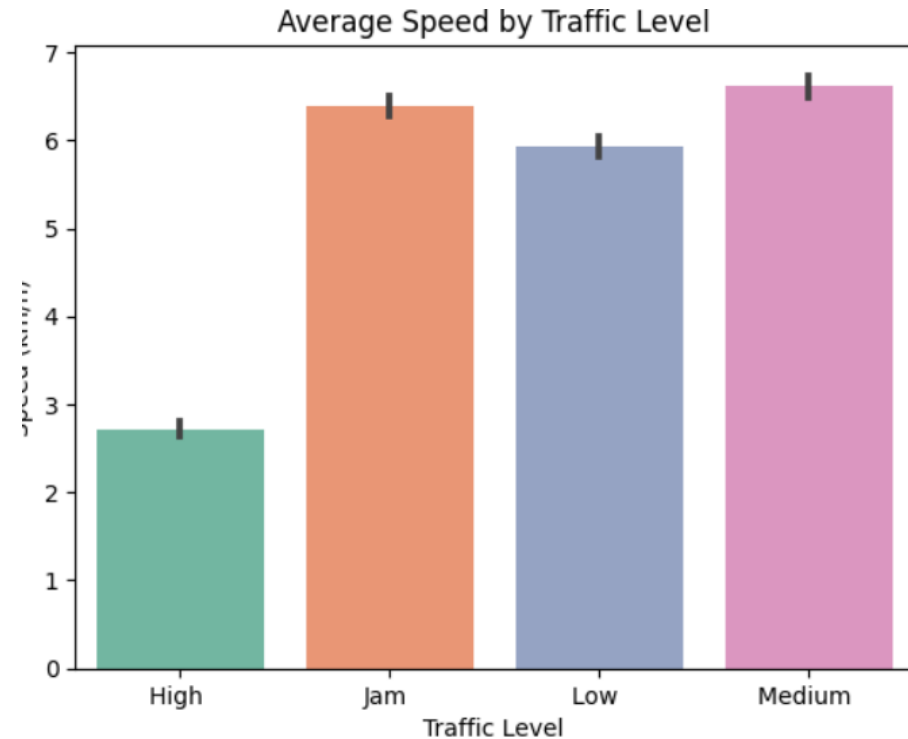


CORRELATION BETWEEN NUMERICAL COLUMN

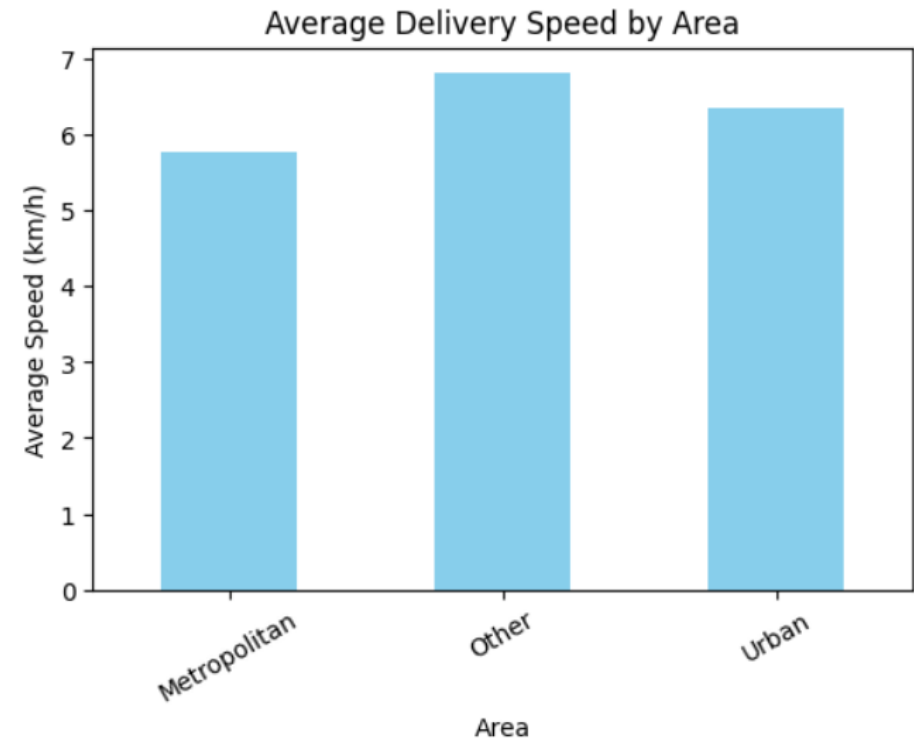


Business Questions they might ask : Analysis Optimization On Delivery speed:

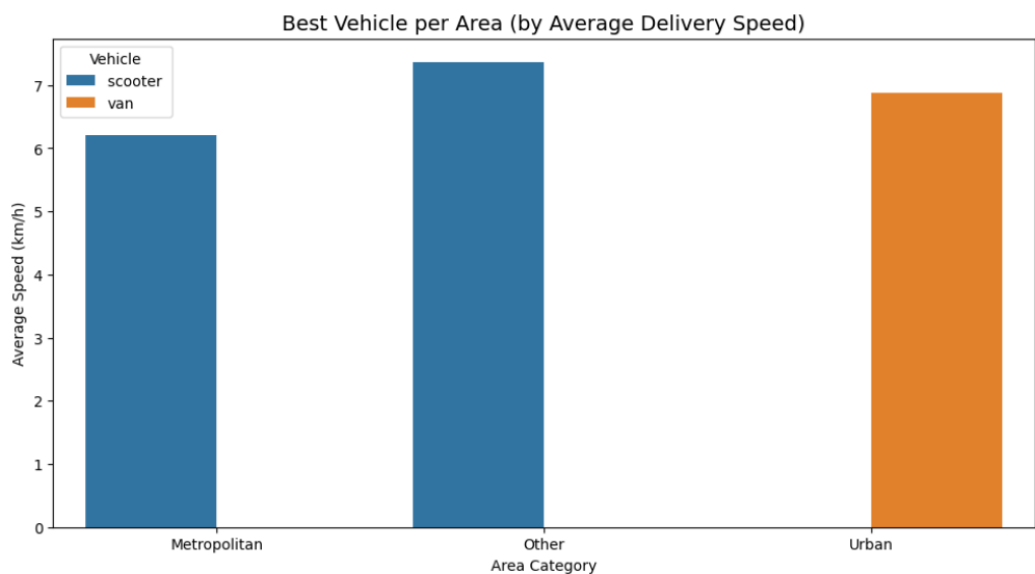
How does traffic impact delivery speed?



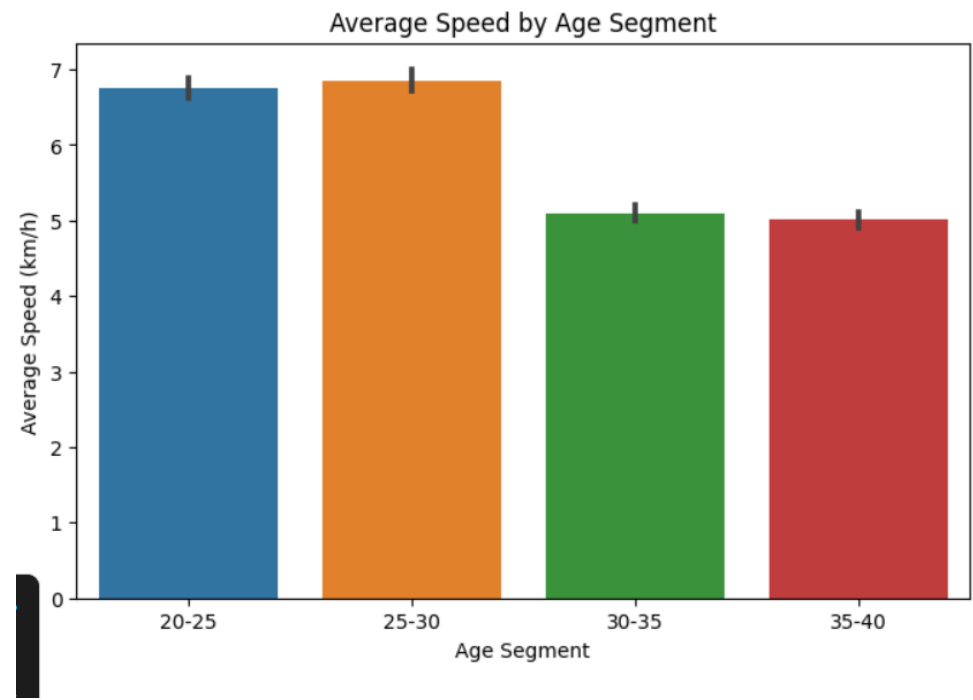
Do area categories affect delivery speed?



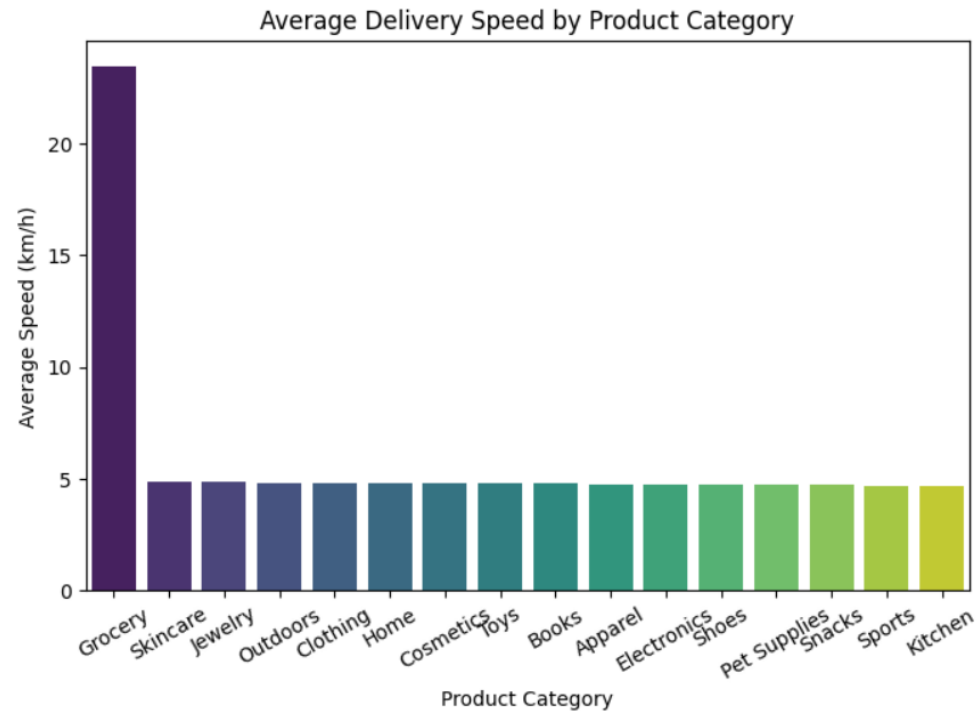
Which type of vehicle achieves the highest average delivery speed in each area category?



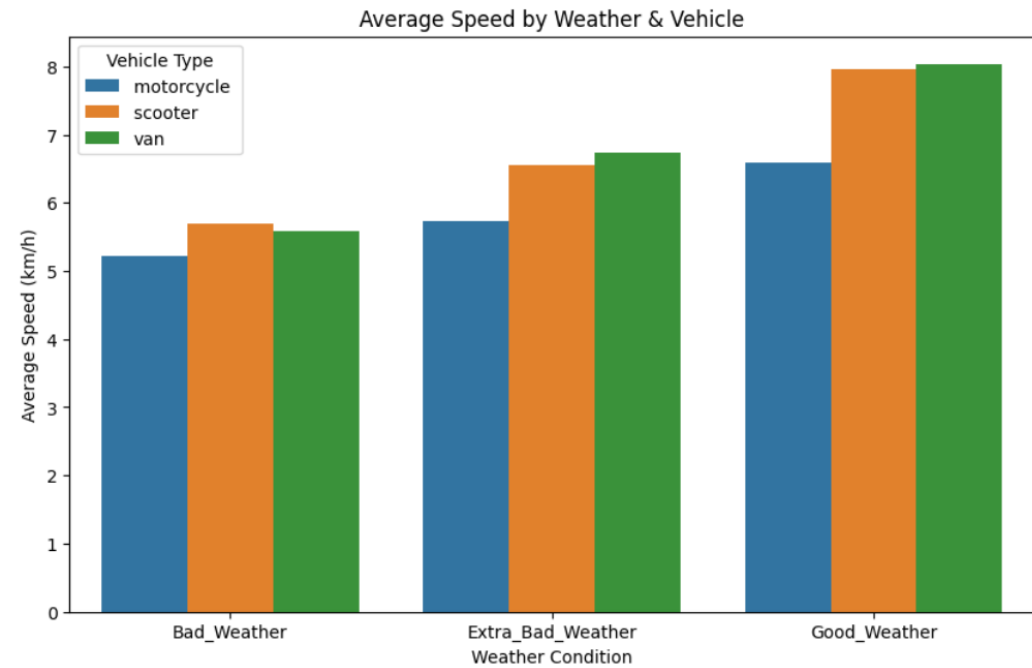
Do age segments affect delivery speed?



Which categories of products are delivered faster?



How do weather conditions and vehicle types together impact delivery speed?



Insights:

- **Traffic Impact on Delivery Speed**

- Higher traffic congestion significantly reduces delivery speed.
- Low and medium traffic conditions result in faster deliveries.

- **Area Categories Effect**

- "Other" areas have the highest average delivery speed.
- Metropolitan areas show moderate performance.
- Urban areas are the slowest, likely due to dense population and road restrictions.

- **Product Categories**

- Groceries have the fastest delivery speeds (20+ km/h), possibly due to priority handling.
- Cosmetics and Toys also perform relatively well.
- Books and Apparel have the lowest delivery speeds.

- **Vehicle Types by Area**

- Scooters perform best in Metropolitan and Other areas.
- Vans are more effective in Urban areas (despite overall slower speed)

- Weather & Vehicle Impact:
 - Good weather boosts delivery speed across all vehicle types.
 - Motorcycles consistently achieve the highest speeds, followed by scooters, then vans.
- **Age Segments:**
 - Younger delivery personnel (20–25, 25–30) achieve higher speeds.
 - Older segments (30–35, 35–40) tend to be slower, suggesting age impacts efficiency.

Recommendations:

- **Traffic-Aware Route Optimization:**

- Invest in AI-driven dynamic routing to avoid high-traffic zones.
- Use real-time traffic monitoring to adjust delivery routes.

- **Area-Specific Strategies:**

- For **Urban areas**, optimize **van routes** and consider micro-hubs to reduce congestion delays.
- For **Metropolitan & Other areas**, focus on expanding **scooter fleets** for faster deliveries.

- **Product Delivery Prioritization:**

- Maintain strong focus on **groceries** as they show highest delivery efficiency.
- Introduce speed-enhancement processes for slower categories like **books and apparel** (e.g., batching, pre-sorting).

- **Weather-Adaptive Policies:**

- On **bad weather days**, prioritize **motorcycles** with trained riders for efficiency.
- Implement safety-first delivery SLAs (slightly longer timelines during storms to balance speed & safety).

- **Workforce Optimization:**

- Recruit and train more delivery riders in the **20–30 age group** for higher performance.
- Provide performance-based incentives to motivate older segments to match efficiency.

- **Fleet & Operations:**

- Expand **scooter and motorcycle fleet** where possible to maximize speed.
- Deploy **vans** strategically in urban congested zones where small vehicles can't carry volume.

- **Overall Strategy:**
- Use **data-driven route planning** (traffic + area).
- Build **area & product-specific delivery policies**.
- Optimize **fleet allocation** (motorcycles/scooters for speed, vans for capacity).
- Focus on **younger workforce segments** while supporting older riders with incentives.